

Computer Network Practice – Appendix Wireshark



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Installation

◆ 링크

❖ <https://www.wireshark.org/download.html>

◆ 설치될 PC의 OS에 맞는 파일 설치

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Download Wireshark

The current stable release of Wireshark is 4.2.0. It supersedes all previous releases. You can also download the latest development release (4.2.0) and documentation.

▼ Stable Release: 4.2.0

- Windows x64 Installer
- Windows Arm64 Installer
- Windows x64 PortableApps®
- macOS Arm Disk Image
- macOS Intel Disk Image
- </> Source Code

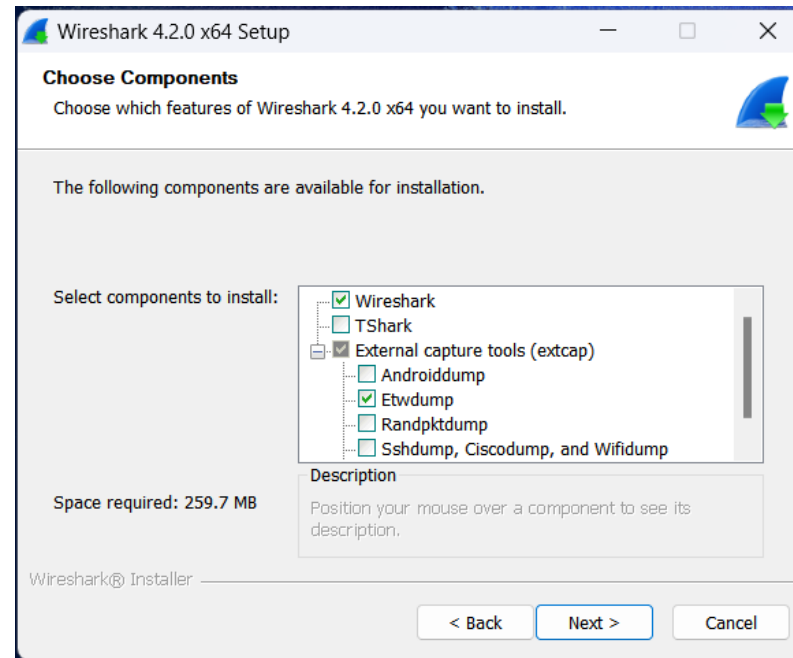
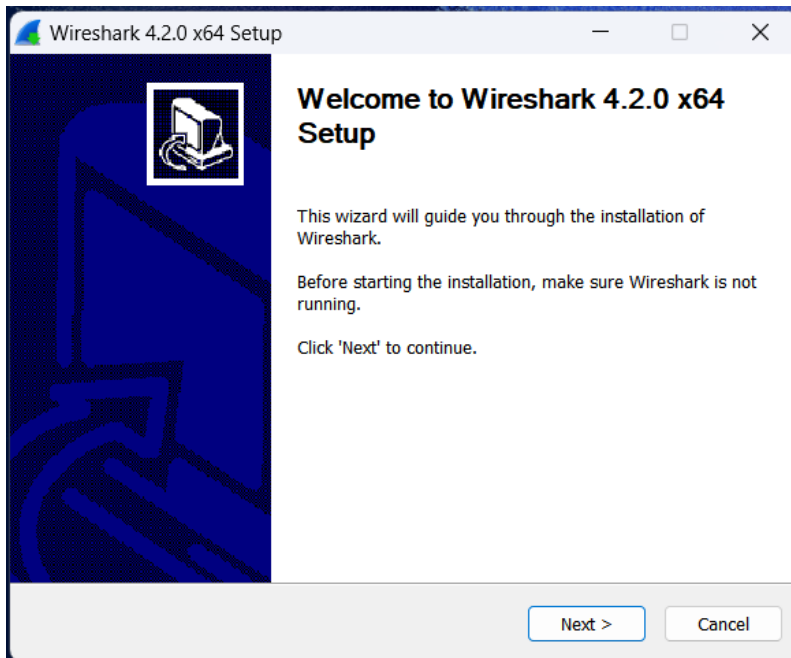


Installation

◆ Wireshark 설치

❖ Tshark 체크 해제

◆ tshark는 CLI 환경에서 wireshark를 이용하는 패키지

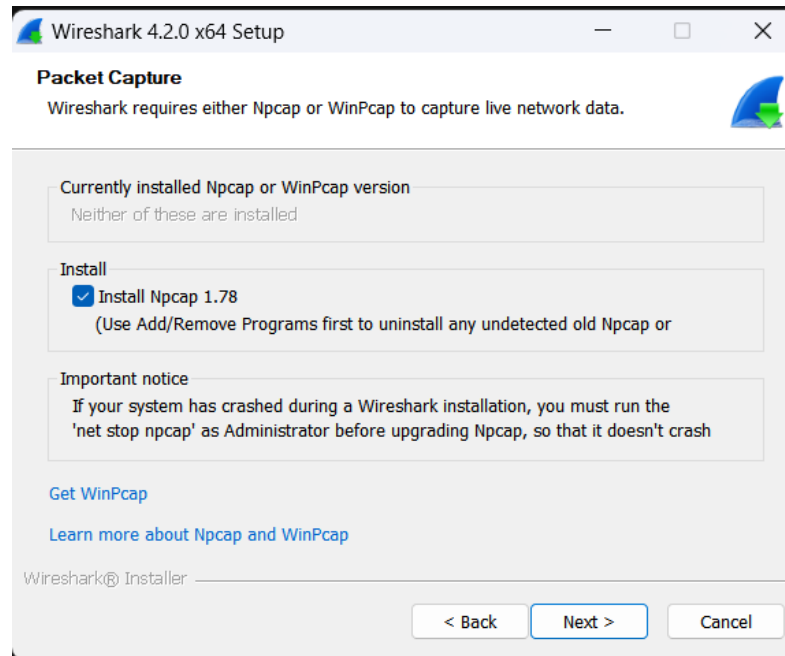




Installation

◆ Winpcap 설치 여부

❖ check하고 Next

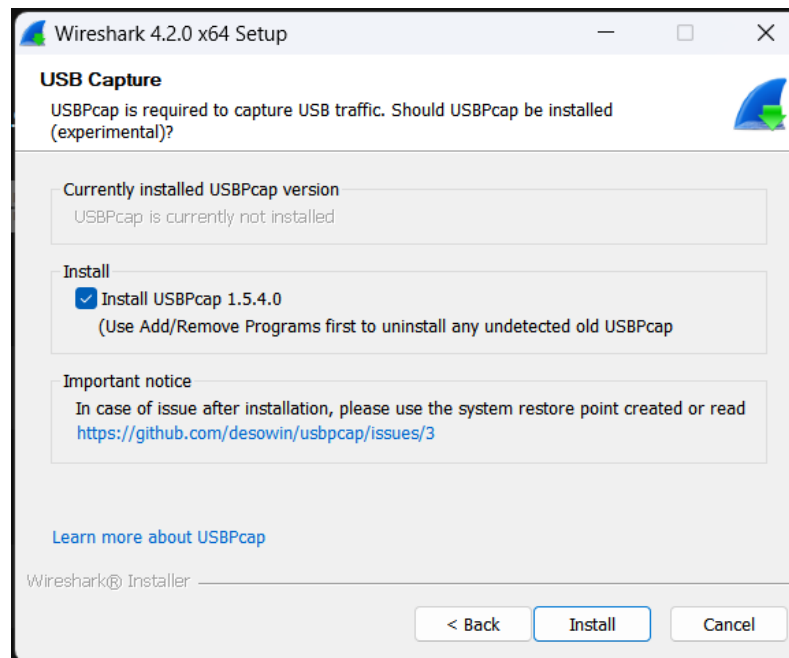




Installation

◆ USBPcap 설치 여부

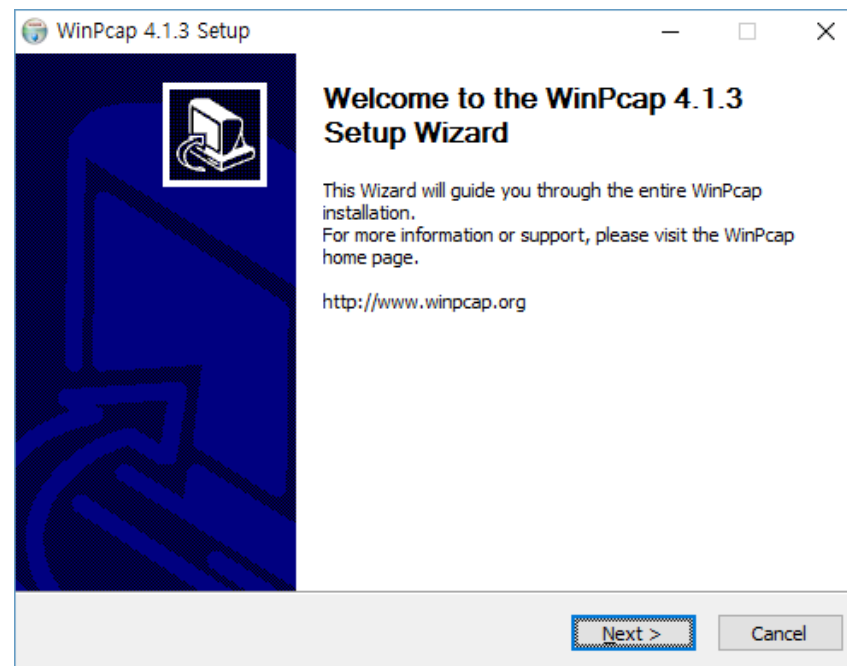
❖ check하고 Next





Installation

◆ Winpcap 설치 화면





How to filter packet

◆ Filtering string

❖ 다음 그림과 같이, 붉은 박스에 filter 명령어를 입력 시 특정 패킷만 리스트로 보여줌

The screenshot shows the Wireshark interface with the filter expression `s1ap and ( sctp.srcport==5603 or sctp.dstport==5603 )` entered in the filter field. The packet list shows several S1AP/NAS-EPS packets. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Stream Control Transmission Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1325...	10161.727516	168.188.127.78	168.188.127.144	S1AP/NAS-EPS	126	id-downlinkNASTransport...
1334...	10276.976953	168.188.127.78	168.188.127.144	S1AP/NAS-EPS	126	id-downlinkNASTransport...
1339...	10315.023422	168.188.127.78	168.188.127.144	S1AP/NAS-EPS	126	id-downlinkNASTransport...
1355...	10450.073838	168.188.127.78	168.188.127.144	S1AP/NAS-EPS	126	id-downlinkNASTransport...
1451...	11441.087316	168.188.127.78	168.188.129.147	S1AP/NAS-EPS	126	id-downlinkNASTransport...
2026...	16592.466036	168.188.127.78	168.188.127.78	S1AP/NAS-EPS	162	id-initialUEMessage, At...
4365...	3799.219527	168.188.128.167	168.188.127.78	S1AP/NAS-EPS	162	id-initialUEMessage, At...
4487...	3928.857244	168.188.128.167	168.188.127.78	S1AP/NAS-EPS	162	id-initialUEMessage, At...
1450...	11430.844086	168.188.129.147	168.188.127.78	S1AP/NAS-EPS	162	id-initialUEMessage, At...
1451...	11441.307079	168.188.129.147	168.188.127.78	S1AP	129	id-initialUEMessage[Mal...

Frame 1450148: 162 bytes on wire (1296 bits), 162 bytes captured (1296 bits) on interface 0
> Ethernet II, Src: Pegatron_2b:c8:4c (84:00:2d:2b:c8:4c), Dst: Tellion_62:14:46 (00:0e:dc:62:14:46)
> Internet Protocol Version 4, Src: 168.188.129.147, Dst: 168.188.127.78
> Stream Control Transmission Protocol, Src Port: 1530 (1530), Dst Port: 5603 (5603)



How to filter packet

◆ Filtering string

- ❖ `ip.dst == 192.168.129.5`
 - ◆ IP header의 목적지 IP 주소가 192.168.129.5인 패킷
- ❖ `ip.src == 192.168.129.5`
 - ◆ IP header의 출발지 IP 주소가 192.168.129.5인 패킷
- ❖ `ip.addr == 192.168.129.5`
 - ◆ IP header의 목적지 IP 또는 출발지 IP 주소가 192.168.129.5인 패킷
- ❖ `tcp.dstport == 128`
 - ◆ TCP header의 목적지 Port 주소가 128인 패킷
- ❖ `tcp.srcport == 128`
 - ◆ TCP header의 출발지 Port 주소가 128인 패킷
- ❖ `eth.type == 0x0806`
 - ◆ Ethernet header의 frame type이 0x0806인 패킷
- ❖ `[eth.src | eth.dst | eth.addr] == D0:A0:3E:00:87:64`
 - ◆ Ethernet header의 [출발지|목적지|둘 중 하나] 주소가 D0:A0:3E:00:87:64인 패킷
- ❖ 이 외 필요한 명령어는 인터넷 검색을 활용

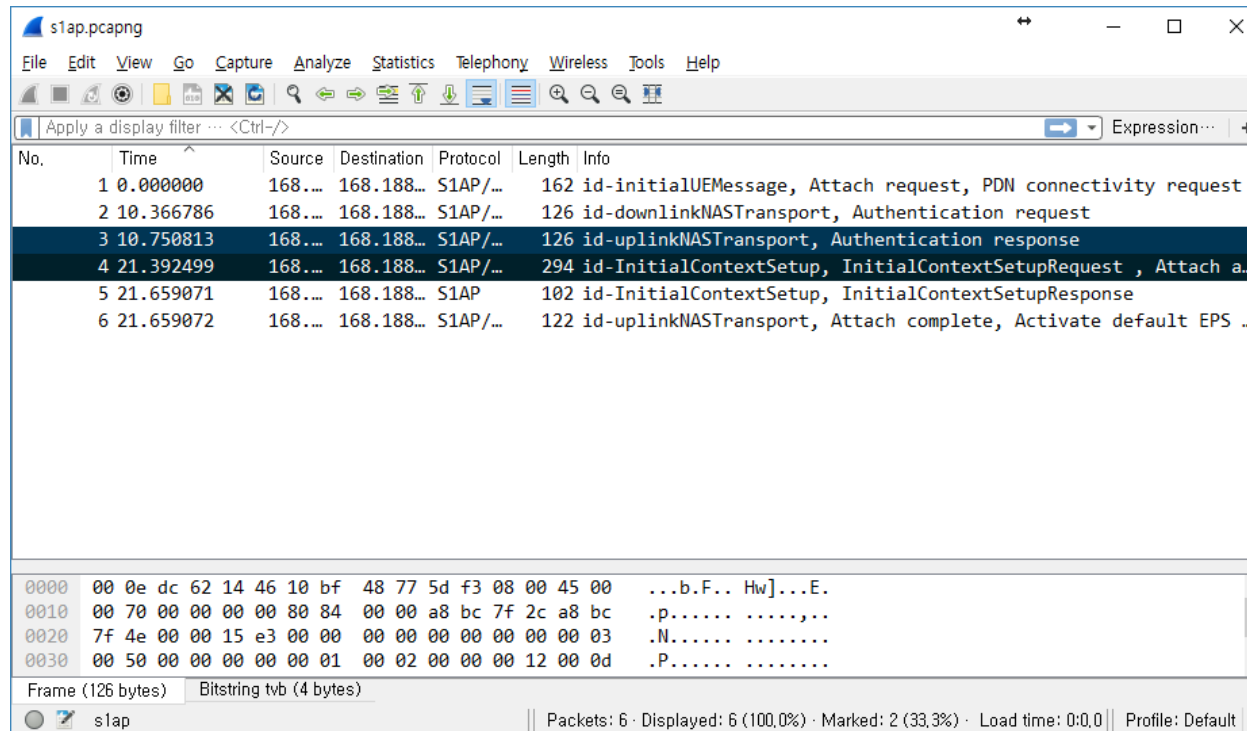


How to export pcap file

◆ Select packets which you export

1. Choose packet using your mouse or keyboard
2. Push the hot key, “Ctrl + M”

◆ As following, the packet mark with a dark color.

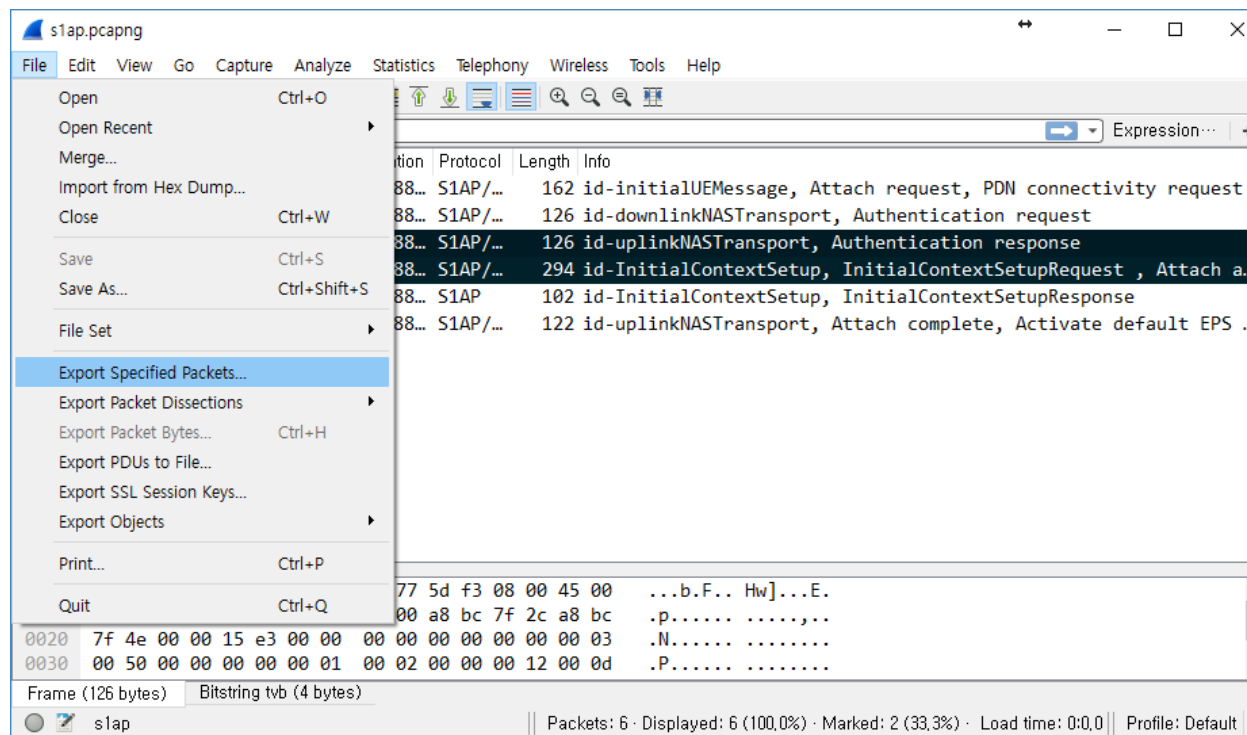




How to export pcap file

◆ Export marked packet

❖ [File] >> [Export Specified Packets...]





How to export pcap file

❖ Export marked packet (cont'd)

- ❖ Check “Marked packets” in [Packet Range Group]
- ❖ Write a file name and Push the save button

